The Wedge Zome: How to Cut and Build It

A nine-sided zome of 36 rhombus panels in 4 rows, 13.8 ft across and 8.7 ft tall, every panel side 35.43 in. It is the Complete Edition's zome, framed with wedges split 8 ways from a 12 in log – the same wedges that frame the dome.

The idea in one paragraph

Every panel is four square sticks – two wedges paired slant to slant and planed 3.87 in square – in a pinwheel, framed flat. Square-edged panels meet on their inside edges and open a V outside of the fold, 14.2° to 27.8°, which a wedge block fills.

What it takes

	this method	the other method
sticks	144	144
wedges, 72 in long	144	72
log sections, split 8 ways	18	9
sticks from one wedge	2	2

Every stick, and how to cut it

Lay the stick **back down** on the saw table with the edge it runs along against the fence. Mitre is the blade's swing on the table, bevel its tilt.

stick	row	how many	length, outside edge	tail: mitre, bevel	head: mitre, bevel
R1A	1	18	28.16 in	57.9° L, 0.0°	57.9° L, 0.0°
R1B	1	18	28.16 in	57.9° R, 0.0°	57.9° R, 0.0°
R2A	2	18	31.08 in	27.3° L, 0.0°	27.3° L, 0.0°
R2B	2	18	31.08 in	27.3° R, 0.0°	27.3° R, 0.0°
R3A	3	18	31.56 in	1.0° L, 0.0°	1.0° L, 0.0°
R3B	3	18	31.56 in	1.0° R, 0.0°	1.0° R, 0.0°
R4A	4	18	31.42 in	15.6° R, 0.0°	15.6° R, 0.0°
R4B	4	18	31.42 in	15.6° L, 0.0°	15.6° L, 0.0°

The rows

row	panels	sharp corner	largest fold	outside V
1	9	32.1°	24.2°	14.2°–24.2°
2	9	62.7°	19.3°	14.2°–19.3°
3	9	89.0°	27.8°	19.3°–27.8°
4	9	74.4°	27.8°	27.8°–27.8°

The wedge blocks

A block's point is 360° divided by the number of ways its log is split, and the V it fills is the fold, so each fold wants a log split 360/fold ways. Mark the split on the end grain with the template for that fold.

fold	seams	split exactly	split	off by
14.2°	18	25.4 ways	25 ways	+0.2°
19.3°	18	18.7 ways	19 ways	-0.3°
24.2°	9	14.9 ways	15 ways	-0.2°
27.8°	18	13.0 ways	13 ways	-0.1°

Order of work

1. **Buck and split.** Buck 18 sections, 72 in long; rip each 8 ways through the pith with the chainsaw, never alone. Trim the backs flat.
2. **Cut the sticks.** 2 sticks from each wedge. Set the bevel once; cut every stick of a row before changing the mitre.
3. **Build the jig, one per row.** Screw stops to a sheet at the row's rhombus outline, full size, from the jig drawing for that row.
4. **Frame the panels.** Four sticks in a pinwheel: each runs to the corner ahead of it and its tail butts the side of the stick behind. Glue and screw at each butt, in the thick part, pilot holes always.
5. **The sill and the rim posts.** The rim zigzags: 18 corners, 9 of them 20.8 in above the sill. Stand a post under each high corner – two quarters back to back make a square post 5.62 in on a side.
6. **Raise, bottom row up.** Fasten panels to each other through the outside seam. Brace the top rows: they sag until the ring closes.
7. **Close the seams.** Fill each outside V with its wedge block, then seal.

Before anyone lives in it

This is a method, not a rated structure. Have a licensed engineer check the frame for wind and snow, and the fastening schedule, before it is occupied.